OCP SPECIAL FOCUS:
PHOTONICS

Open Optical Solutions for AI Scaling

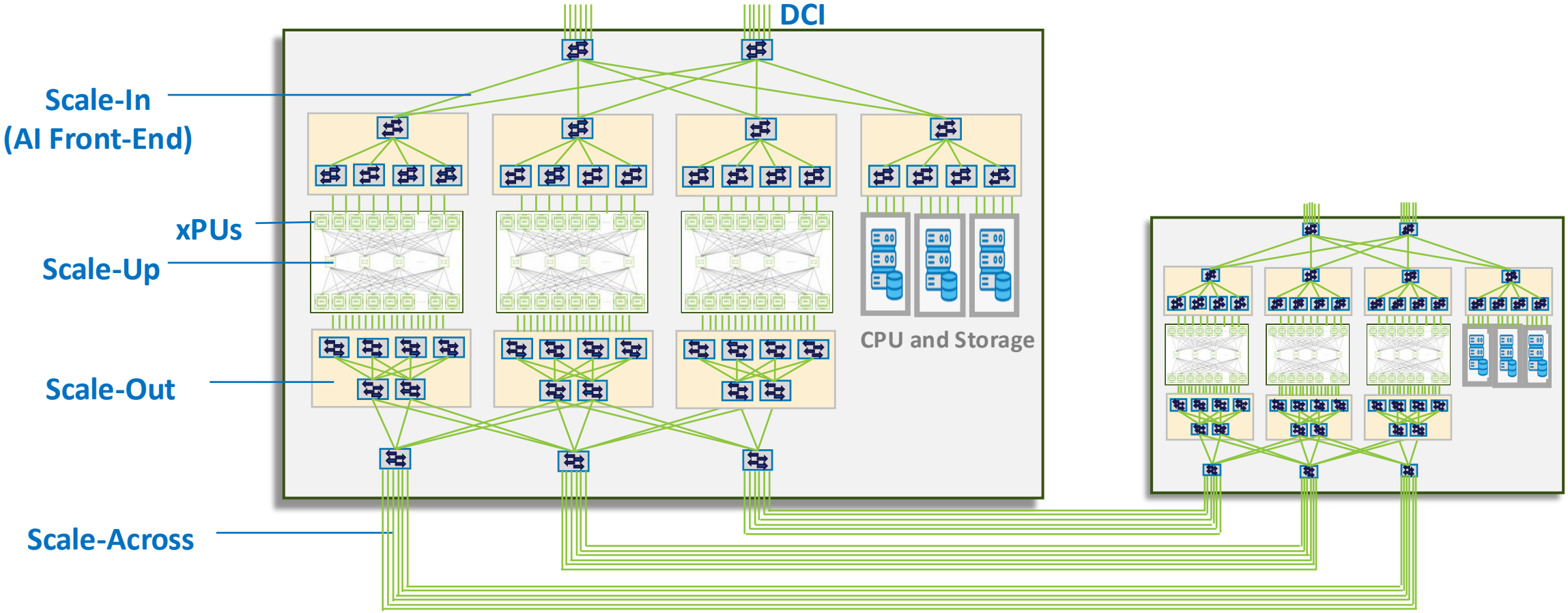
Rang-Chen (Ryan) Yu

VP Marketing, TeraHop

OCP SILVER Member

Optics Connectivity Challenges for AI Scaling

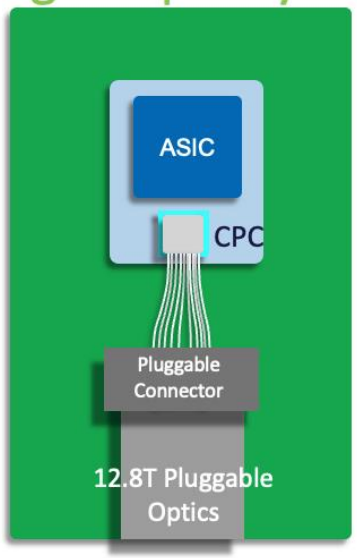
- Optical connectivity solutions required to support AI scaling from Scale Up, Scale out and Scale Across
- AI scale up transitioning from copper to optics with opportunity and challenges
- Open optical solutions critical to support AI infra demand and growth



2025 OCP Call for Open Optics for AI

- Called for industry to work together to provide open optics supporting high capacity pluggable and Open Socket NPOs

High-Capacity Pluggable Optics for Scale Up

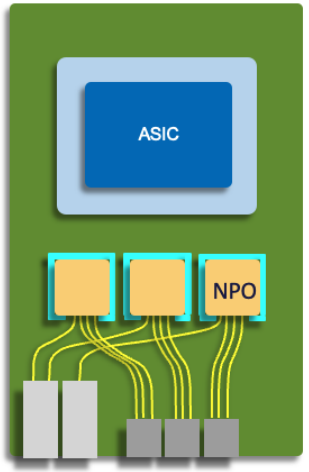


- Scale-up BW needs ~ 10x more than scale-out
- No intrinsic limit to high channel count optics in one single PLO
- e.g., OIF Hi-Density connector project
  - New High Density 2D Electrical Connector (beyond OSFP) parameter example:
    - BW: 12.8T / 25.6T
    - Pitch: 0.5mm
    - High lane count: 64 Tx/Rx signal pairs + 16 power & control
    - Design I/O rate: 224G PAM4 / 448G PAM6 or PAM8 (TBD)
    - Size: 2x OSFP W (50mm), 1.5x OSFP H (15mm), 1x OSFP L (110mm)

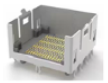


- 12.8T Transceiver:
  - High capacity/high density PLO
  - Compact size for GPUs and Scale-up switches
  - Improved reliability with silicon photonics
  - Lower $/Gbps with higher degree integration

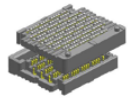
Open-Socket NPO



- Sustaining long developed open optical supply eco system
- Many type of deployed high-speed connectors
  - Rated to support 200G-PAM4 per lane
  - Many deployed for copper cable applications already
  - DP density from 0.2 to 2 DP/mm^2


AdrenaLINE (TE)


SiFly (Samtec)


MirrorMezz (Molex)


Paladin HD2 (Amphenol)

- Open Socket enabling open eco system like PLO
- Applications/Use Cases
  - GPU/ASIC server scale up
  - Switch for Scale Up
  - BE NIC card optics

OCP GLOBAL SUMMIT | 2025 Leading the Future of AI.

OCP GLOBAL SUMMIT | 2025 Leading the Future of AI.

Rang-Che Yu, "Optical Technology Considerations for AI xPU Scale Up", OCP Global Summit 2025

OCP GLOBAL SUMMIT | 2026 Scaling Innovation for the AI Era.

2026 OCP Global Summit

XPO
OPTICS INNOVATION

ARISTA Amphenol ciena TERAHOP
COHERENT eoptolink MARVELL LUMENTUM
SP SOURCE PHOTONICS Genuine optics LIGENT molex Accelink



- 12.8T XPO MSA announced at OFC’2026
  - Highest capacity and density pluggable
- Product demo at OFC’2026
- 100+ Companies joined
- Release 1.0 spec. release July 31’2026

OPEN CPX MSA

ciena COHERENT MARVELL molex
samtec TERAHOP

Accton ACON OPTICS aoPT ALPHA Networks Amphenol AOI AsteraLabs AVICENA BizLink ColorChip
CREDO eoptolink FIT Genuine optics intel LIGENT LIGHTMATTER Linktel LINTES LOTES
LUMENTUM LYNTERA MIXX multiLane muRata nexthop ai NVISION Photonics
Qualcomm Dragonfly Ruijie RVI SILUXTEK SP SOURCE PHOTONICS TE TFC upscale VIAVI

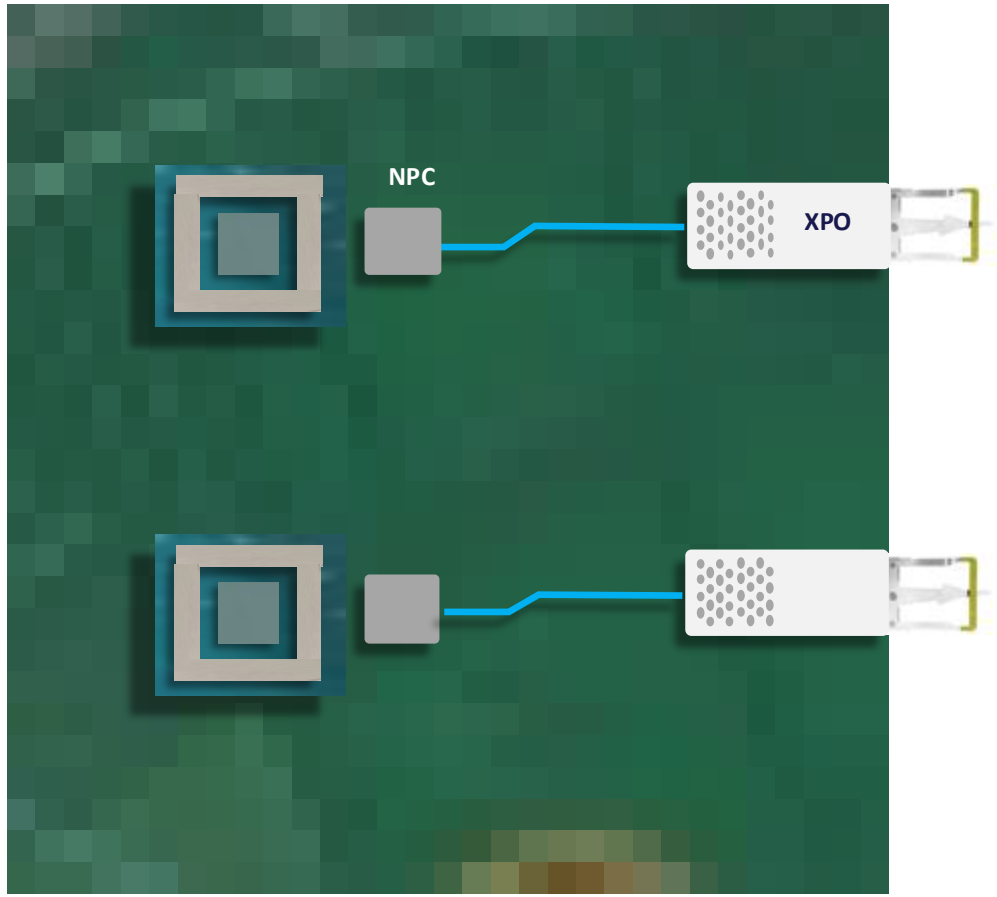


- Open CPX MSA announced at OFC’2026
  - Open socket CPO and NPO engine
- Product demo at OFC/ECOC 2026
- 50+ Companies joined
- Release 1.0 spec. release September 17’2026

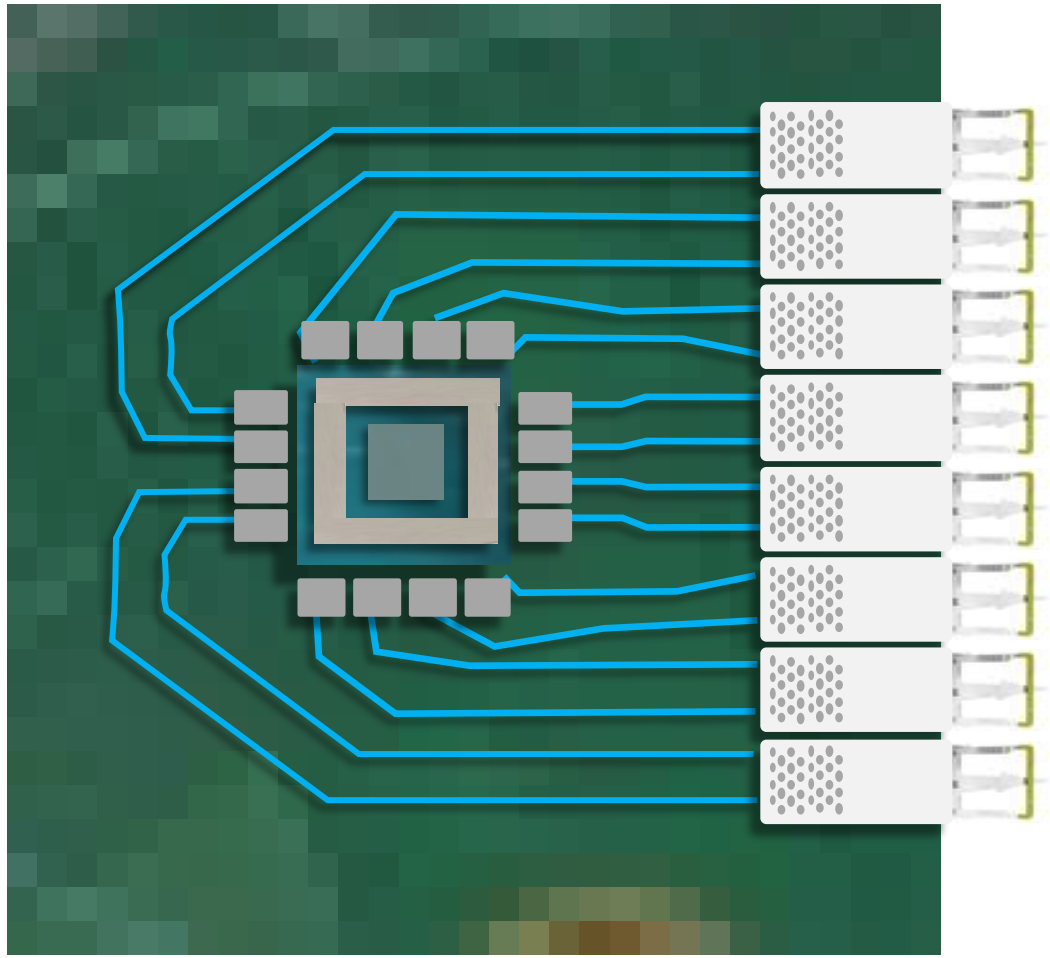
Use Cases of XPO

XPO
OPTICS INNOVATION

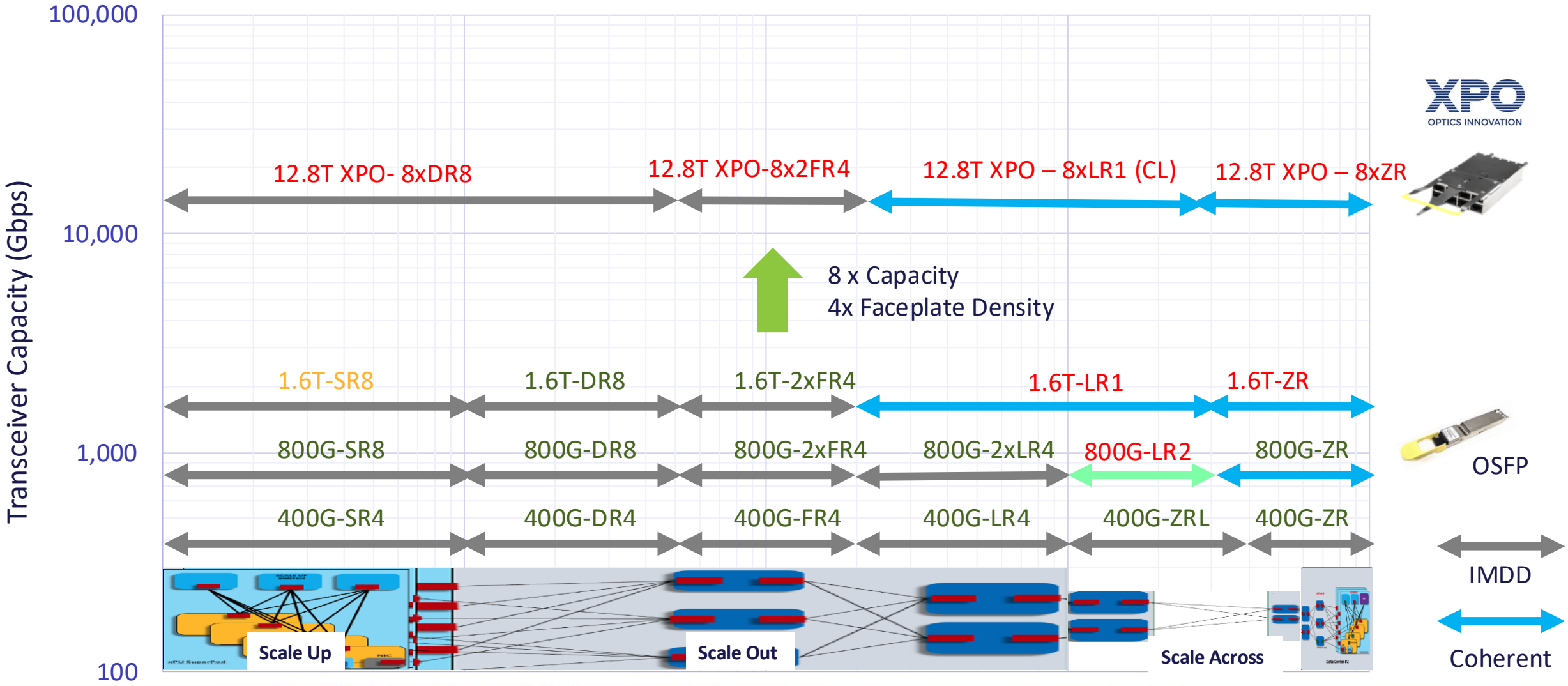
xPU optical interface for Scale Up



High-Capacity Switch Interface for Scale Up/Scale Out



XPO Supports Full Range of AI Scaling



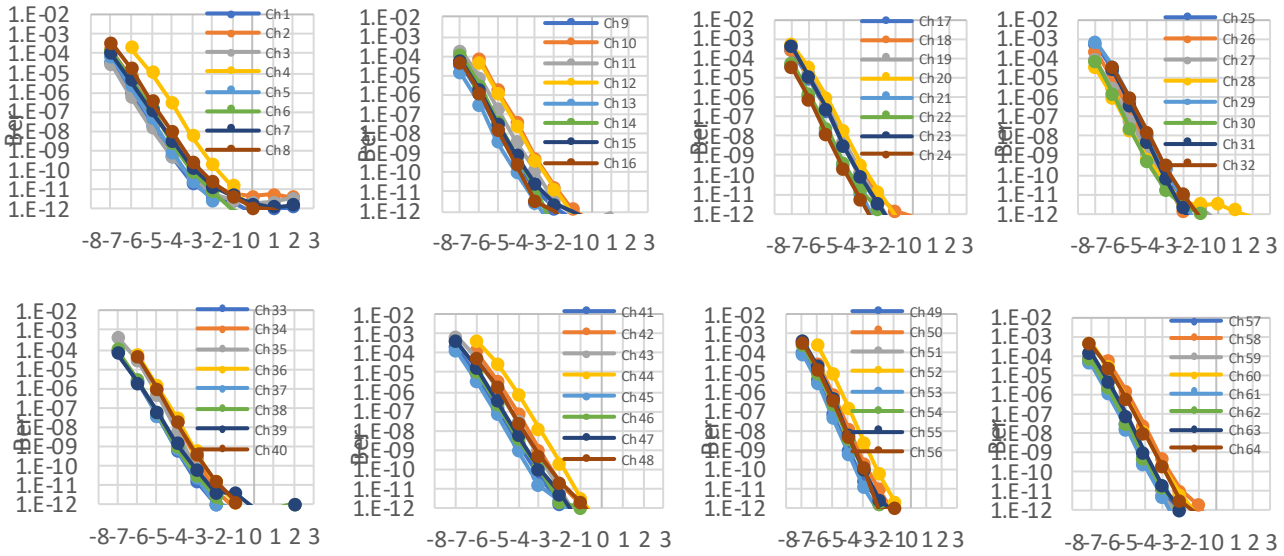
TeraHop 12.8T XPO-FRO Performance

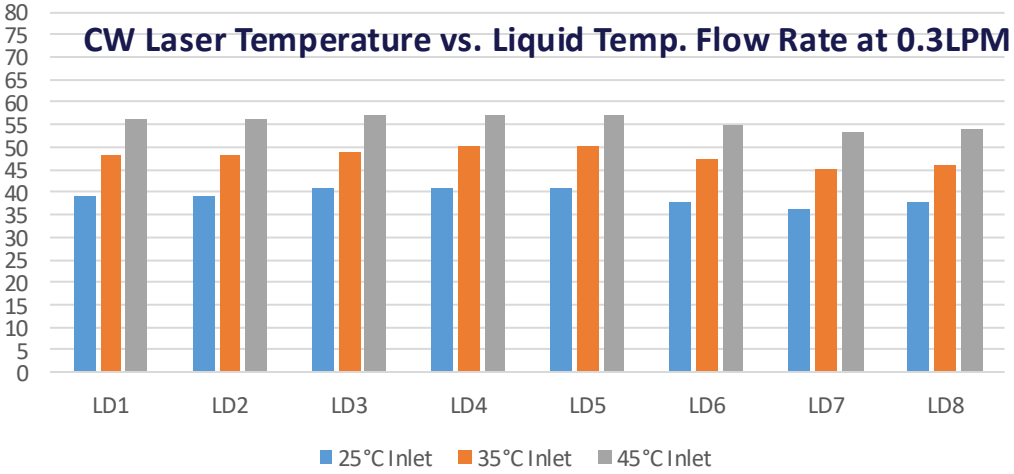


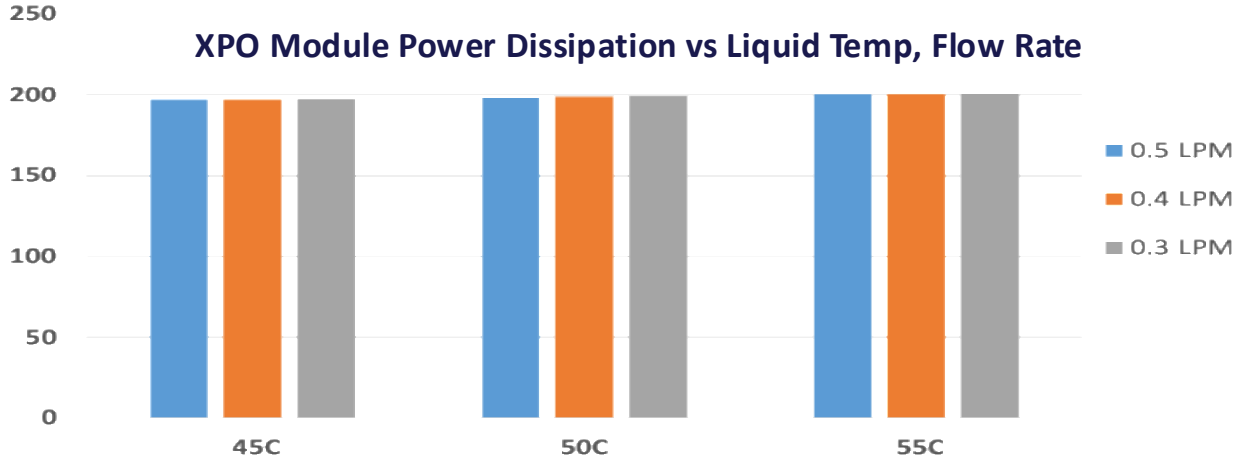
64 x Transmitters



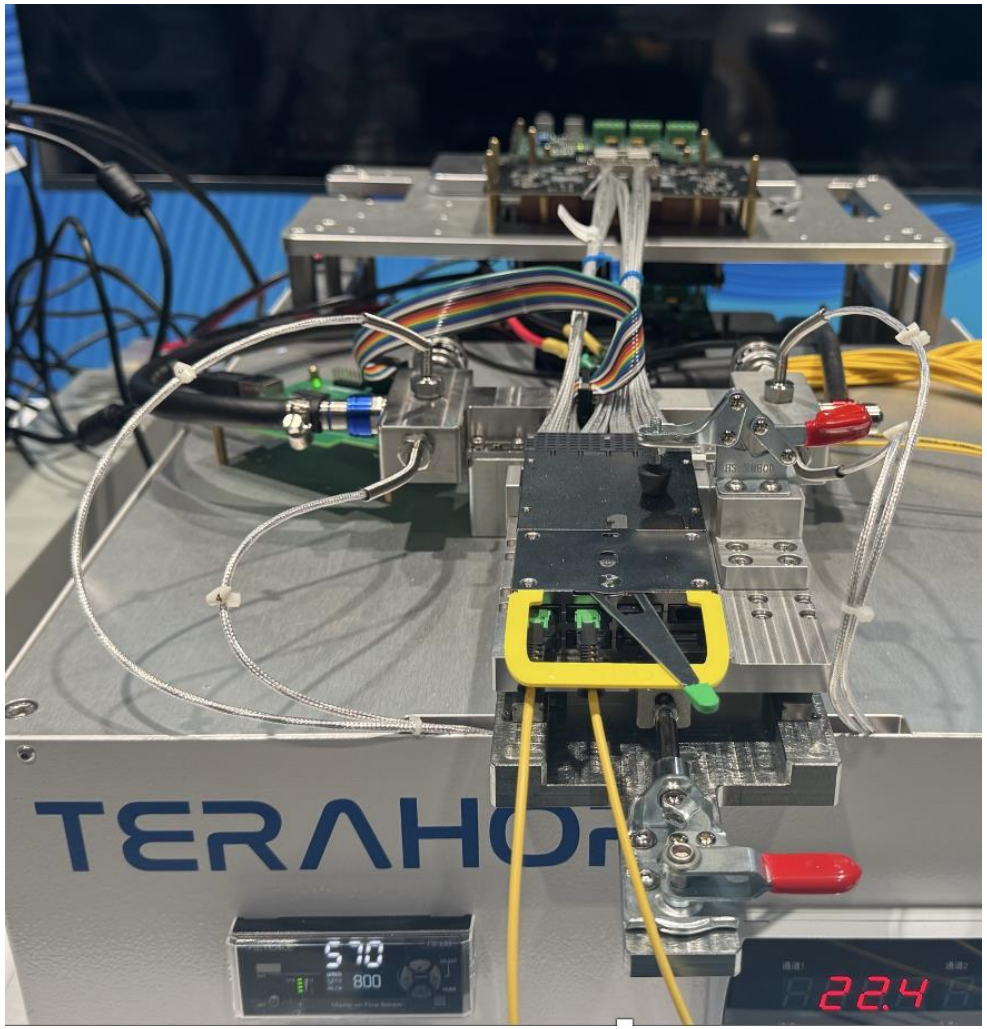
8 x 1.6T-DR8 Link Pre-FEC BER







TeraHop 12.8T XPO-RTLR Demo (Booth # A32)

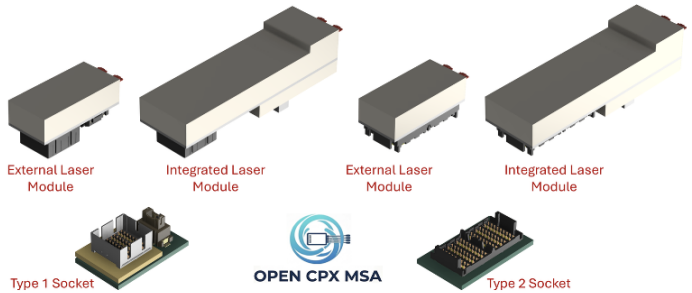


CH	EdStart	ErrCnt	BitCnt	_Ber	R_Lock	L_Lock
1		2E+0	2.690463E+12	7.43367E-13		
2		8E+0	2.655825E+12	3.01225E-12		
3		1E+0	2.620975E+12	3.81537E-13		
4		8E+0	2.589313E+12	3.08962E-12		
5		1.2E+1	2.5568E+12	4.69337E-12		
6		1E+1	2.52365E+12	3.96251E-12		
7		4E+1	2.490075E+12	1.60638E-11		
8		6E+0	2.457775E+12	2.44123E-12		
9		0E+0	2.423988E+12	0E+0		
10		5E+0	2.38935E+12	2.09262E-12		
11		1E+0	2.354288E+12	4.24757E-13		
12		3.09E+2	2.31965E+12	1.3321E-10		
13		2.5E+1	2.287138E+12	1.09307E-11		
14		4.6E+1	2.256113E+12	2.03891E-11		
15		5.4E+1	2.221688E+12	2.43058E-11		
16		1.4E+1	2.191725E+12	6.38766E-12		
17		1.958901E+8	2.159638E+12	1E+0		

Est. Channel Loss ~ 22dB

Open CPX MSA Roadmap

Open CPX MSA Roadmap



Publications

September
Rev 1.0 Spec: 32/36 x 200G
Type 1 & 2, ILM & ELM

March
Rev 1.1 Spec: 32/36 x 200G
Copper media, Type 3, other TBD

June
Rev 2.0 Spec: 64/72 x 200G
Socket type(s), ILM & ELM, Copper

Late 2027 / Early 2028
Rev 3.0 Spec: N x 400G

2026 — 2027 — OFC — 2028

Milestones

March
Open CPX MSA Founded

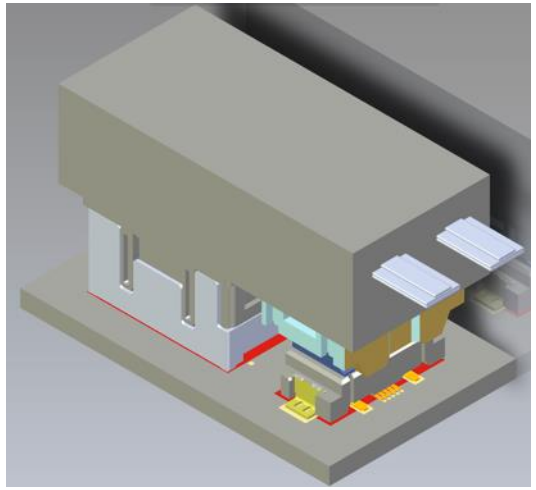
September
Open CPX Tops 50 Members

OPEN CPX MSA

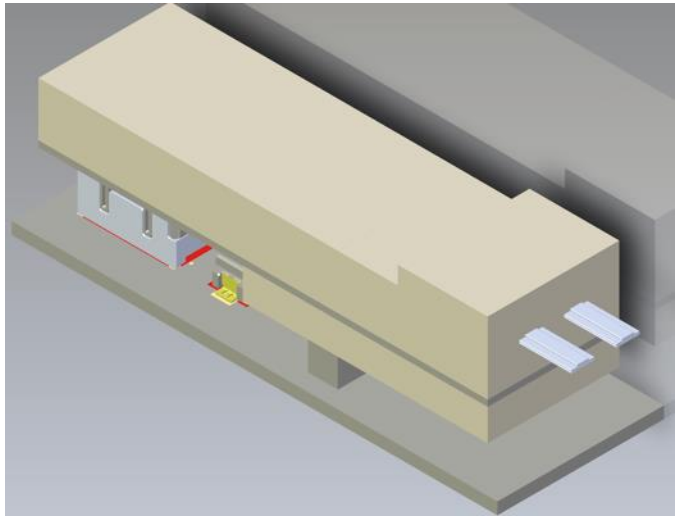
Subject to change without notice.

Open CPX MSA Release 1.0

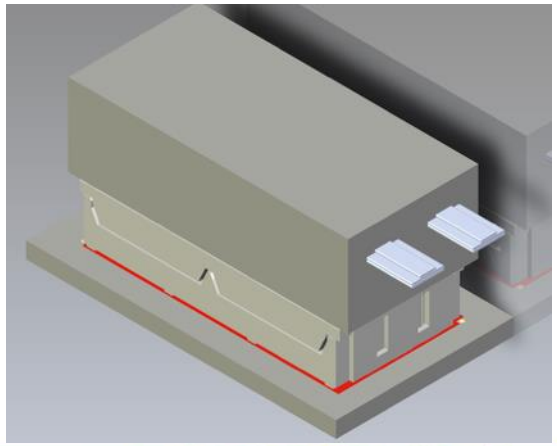
Type 1 Connector, ELM



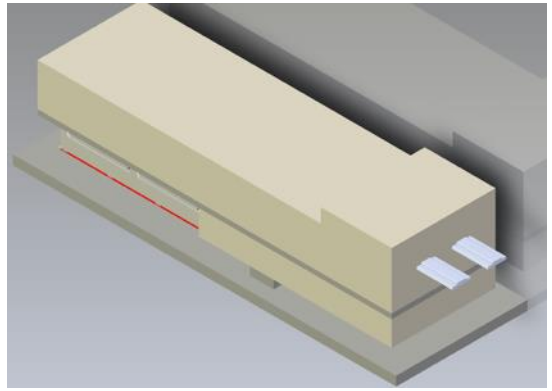
Type 1 Connector, ILM



Type 2 Connector, ELM

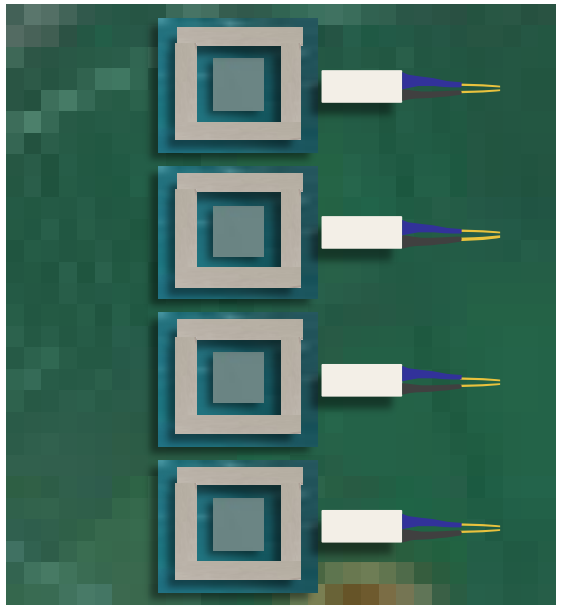


Type 2 Connector, ILM

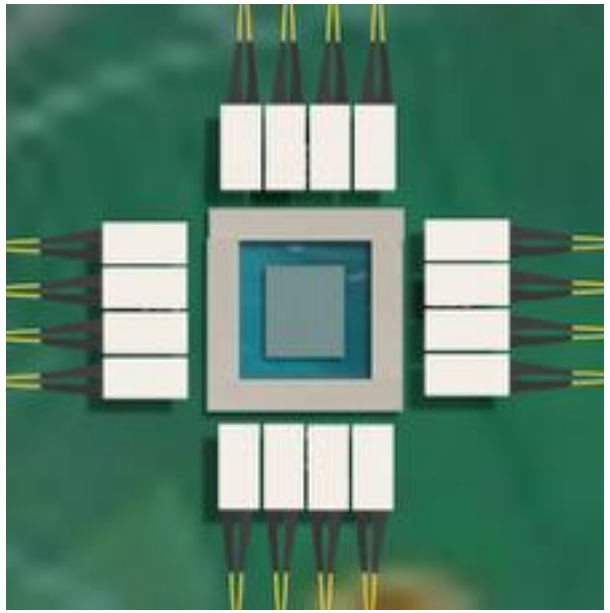


Open CPX Optics Use Cases

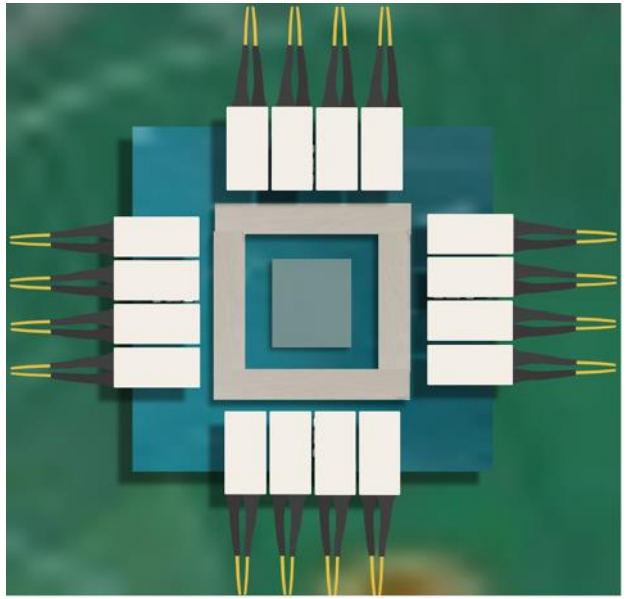
NPO for xPU Scale up



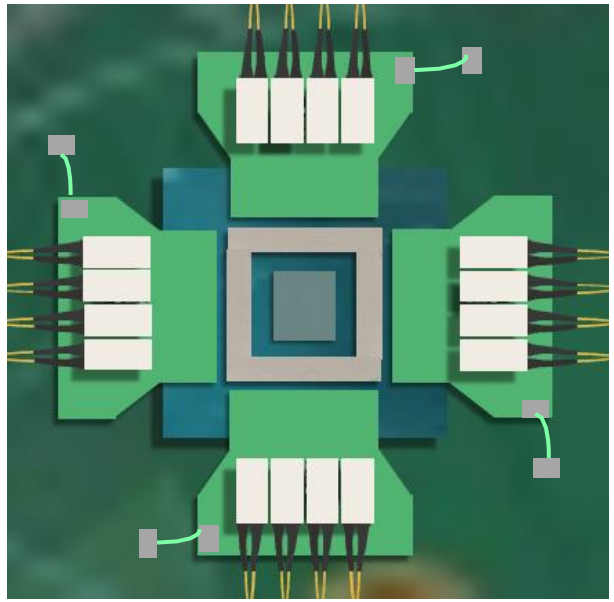
NPO for Switch Scale up/Out



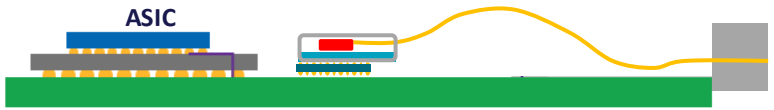
CPO for Switch Scale up/Out



IOA (Integrated Optical Attachment)



Open CPX NPO



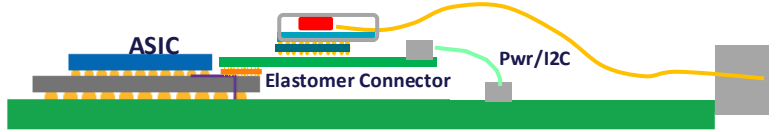
- Optical Engine on Host PCB
- Type 1 or Type 2 Connectors

Open CPX CPO



- Optical Engine on ASIC substrate
- Type 1 connector with high speed DP on ASIC substrate, power/I2C on host PCB

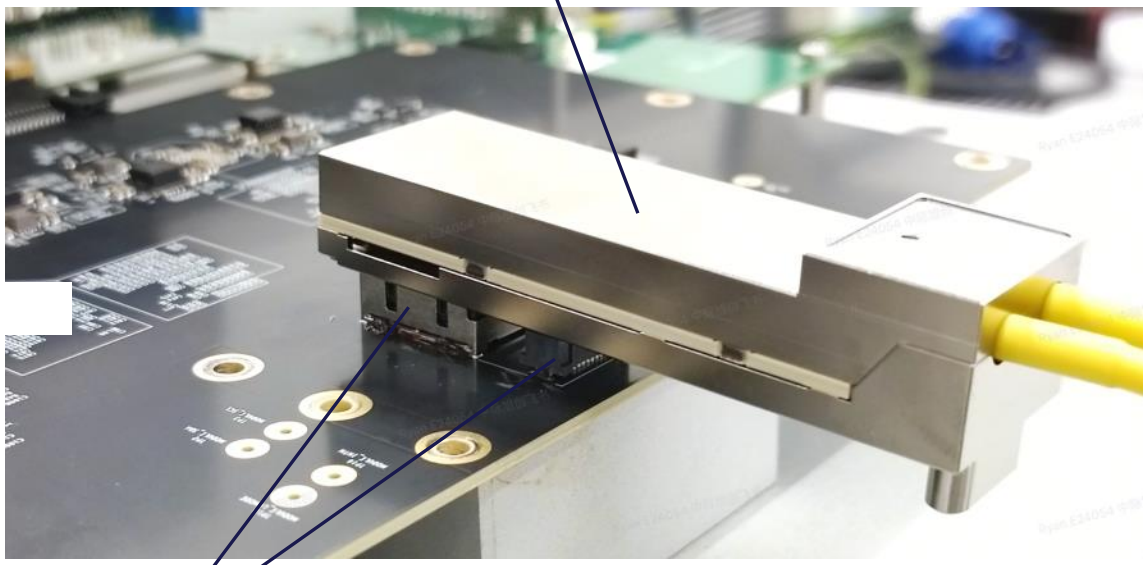
Open CPX IOA



- Optical engine on HDI board attach to ASIC substrate via elastomer high density, high speed low loss connector
- Type 1 or type 2 connectors

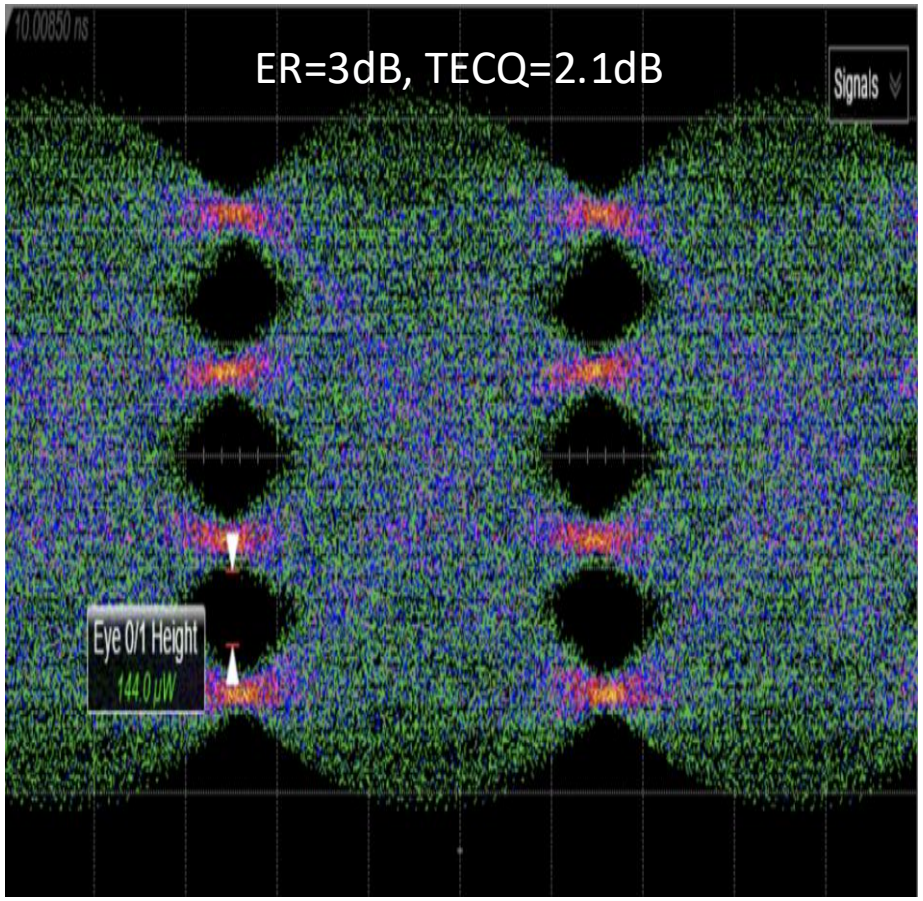
TeraHop “Diablo-1” 6.4T Open CPX Engine

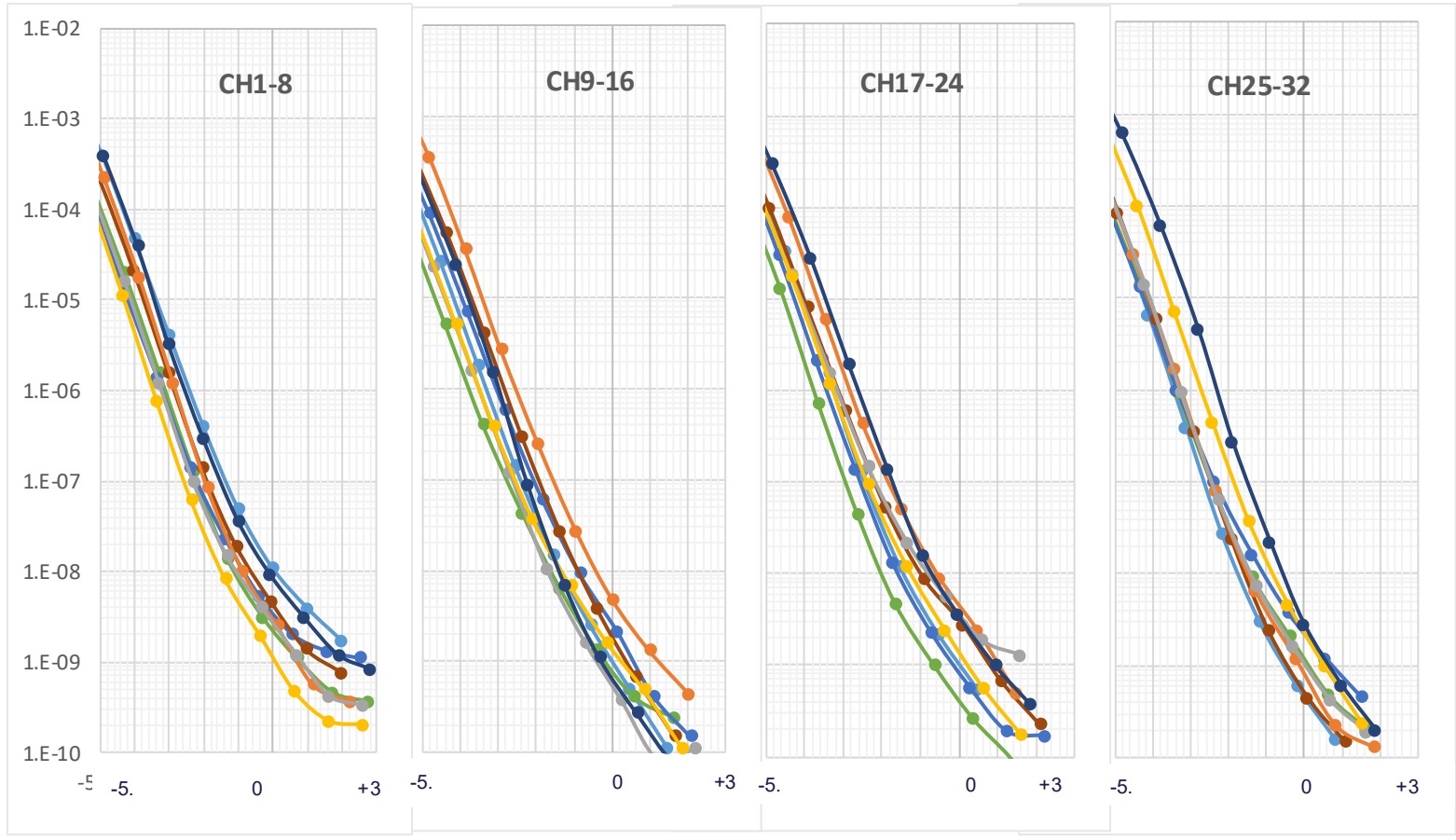
Silicon Photonics Optical Engine with Integrated laser Source



Open CPX Type 1 Connectors

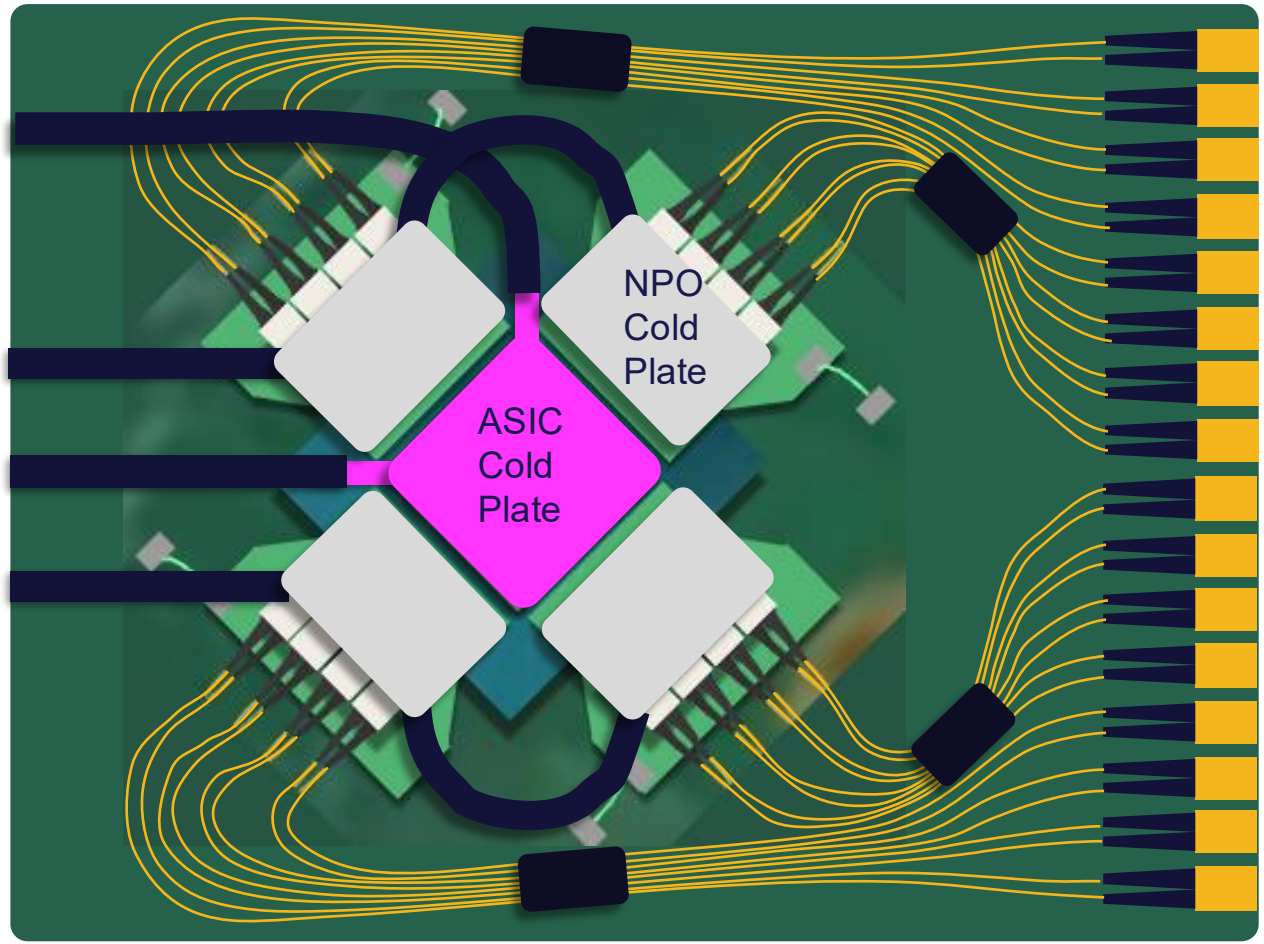
TeraHop “Diablo-1” 6.4T NPO Prelim. Test Data





* Insertion loss between ASIC to NPO ~ 12dB

“Diablo-1” 6.4T NPO Key Attributes



Integrated laser with proven low FIT

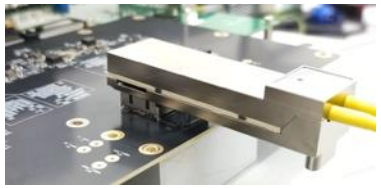
- NPO close to, but separate from ASIC
  - Separate cold plate from ASIC
  - Critical components including LD at modest temperature
- Low power ~ 5.5pj/b, including internal laser
- Simple SMF pigtail termination to front panel
  - No extra connector loss and potential failure

ELSFP disadvantage

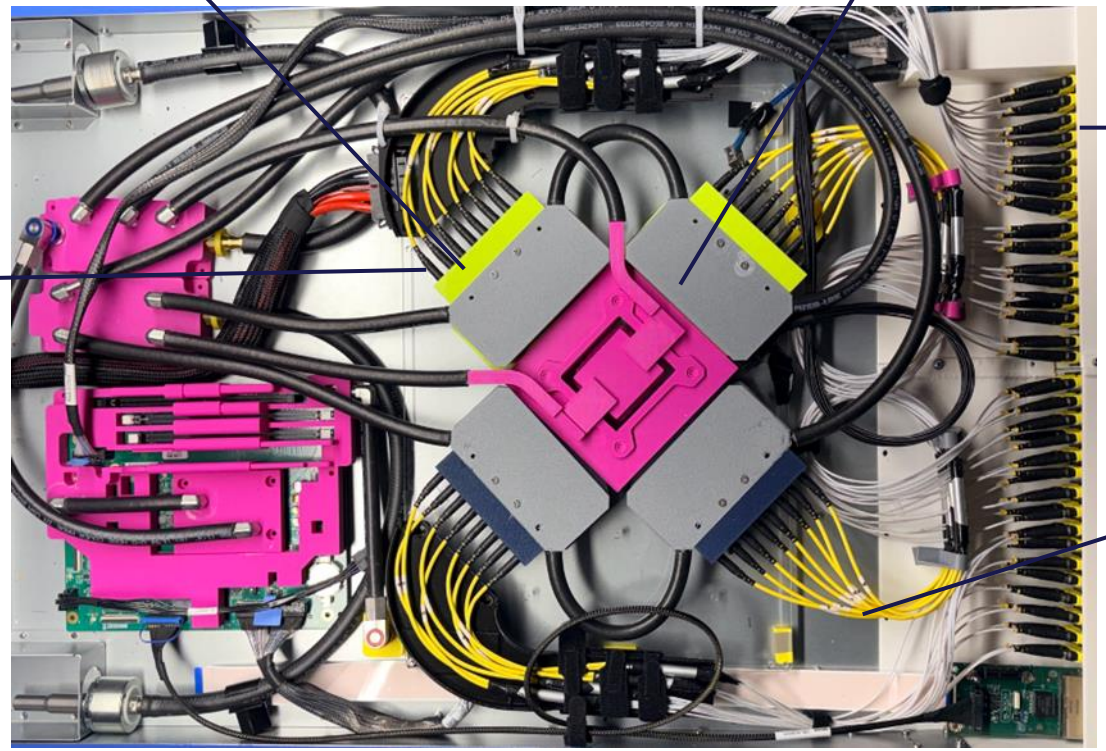
- lose > extra 3dB power
- →requires > 2x CW LD optical power (200-300mW+)
  - increase CW LD power dissipation by ~ 4x with TEC
- PM fiber adding additional power loss and cost
- Blind-mate connector adding loss and failure
- Higher cost, power
- Lacking field reliability data

100T Switch Example with “Diablo-1” NPO

- “Diablo-1” 6.4T NPO with integrated lasers,



- Integrated fiber pigtail, no failure-prone optical interface

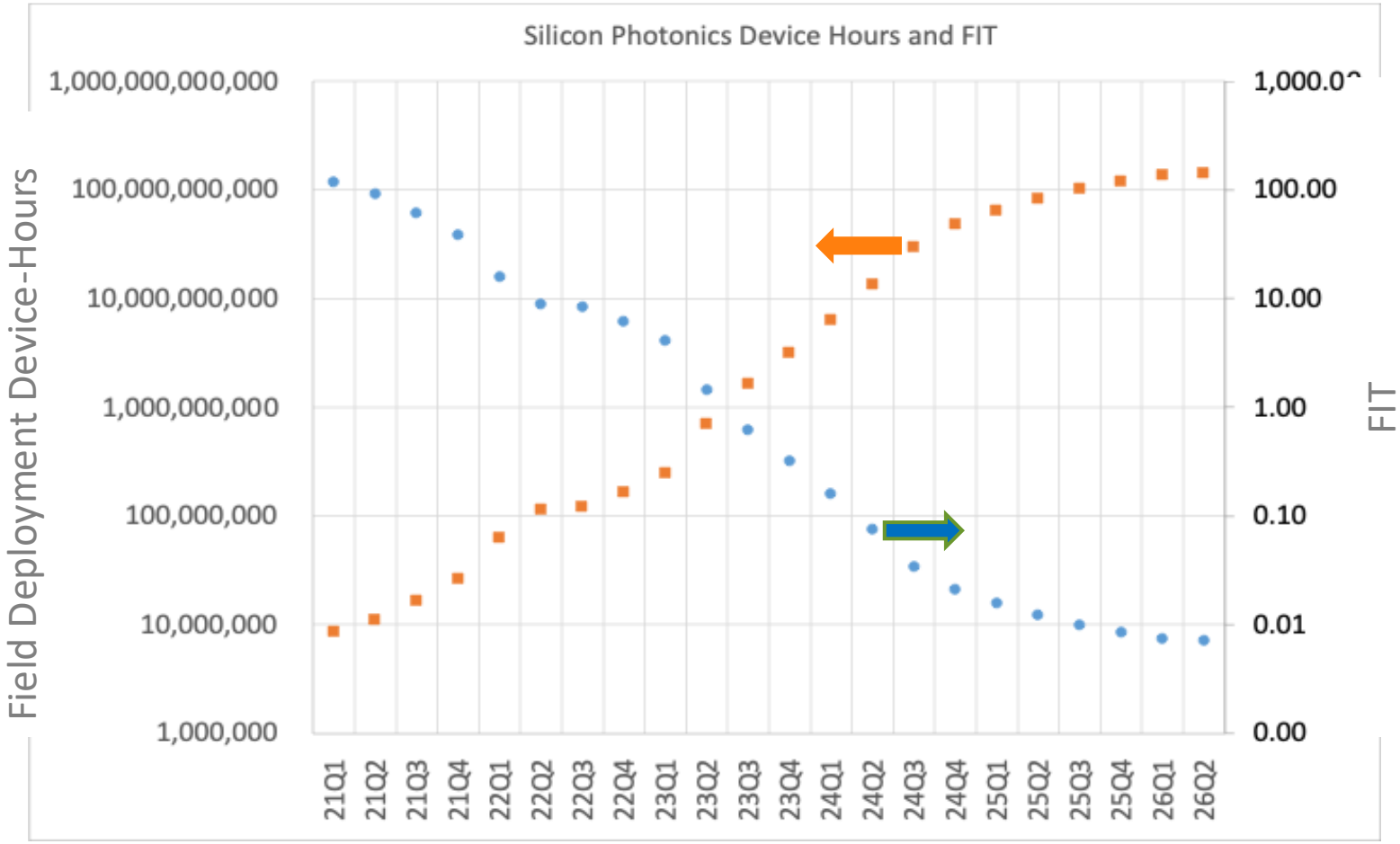


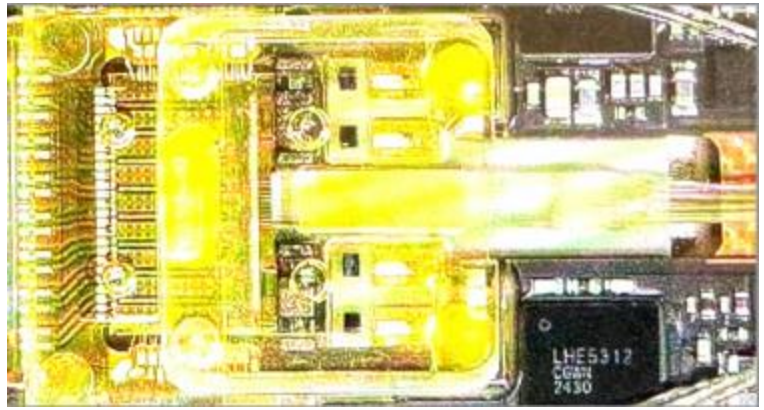
- Dedicated Liquid-Cooling Cold-plate for NPO, keep all critical components cool
- Clean Front Panel with optical I/Os
  - No ELSFPs taking up large front panel space
- Clean SMF fiber bundles to the front panel
  - No extra PM fiber from ELSFPs

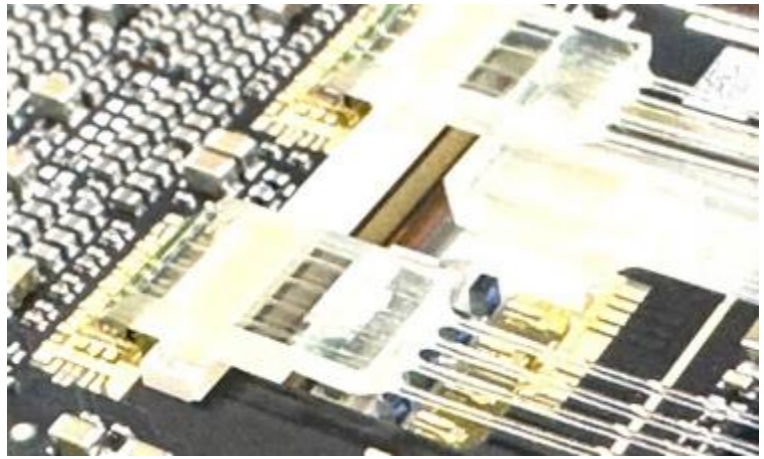
Industry 1st 100Tbps switch Design-Concept by **Arista**, with TeraHop 6.4T “Diablo-1” Open CPX Engine, with permission by Arista

Silicon Photonics PIC Reliability Discussions

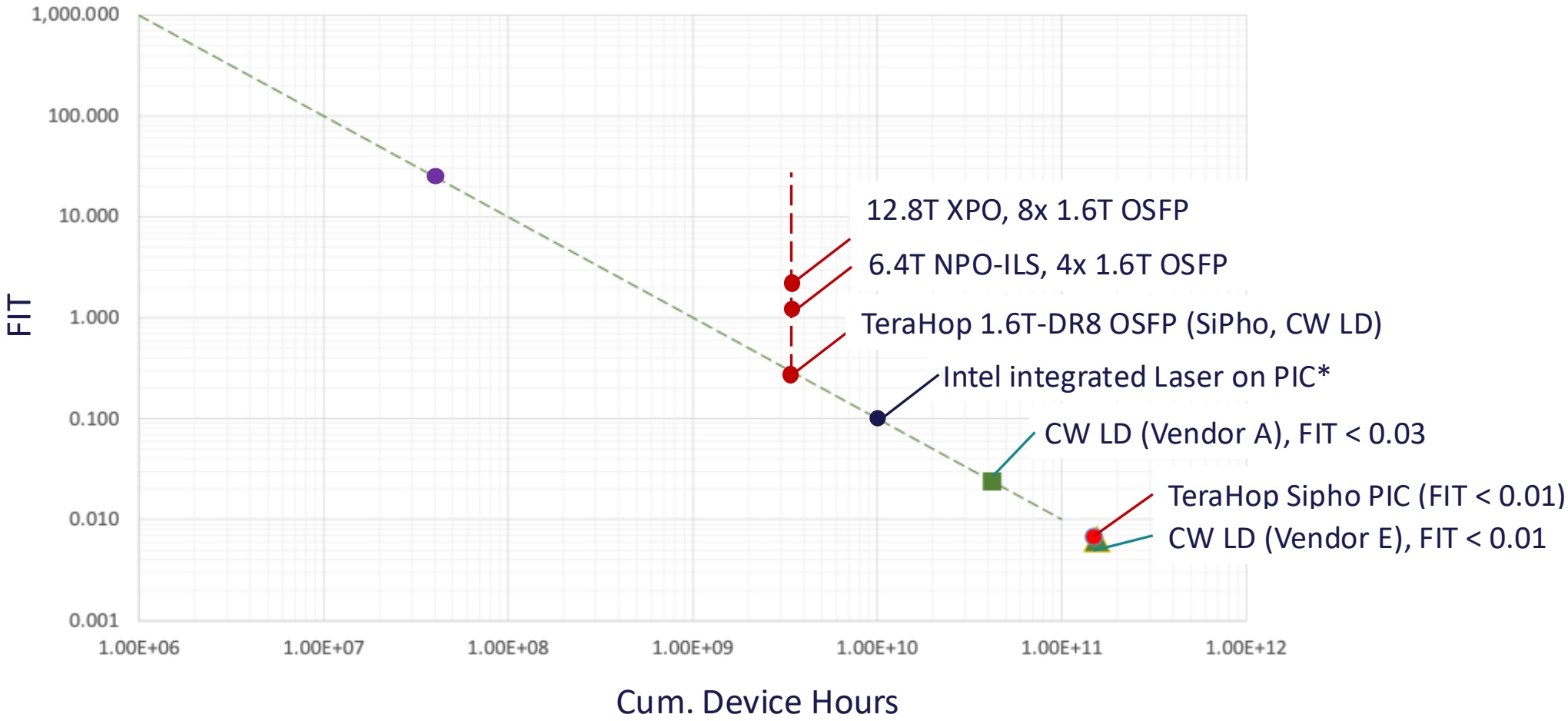
- Not all “silicon photonics” created equal: maturity and high rel. needs field data
- TeraHop shipped Silicon Photonics optical transceivers with PIC >100B Device-Hours, FIT< 0.01







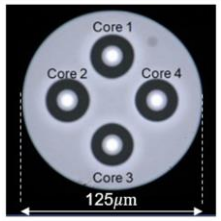
XPO and Open CPX Engine Reliability Discussion



* Source: Intel. https://www.intel.com/content/www/us/en/newsroom/news/intel-unveils-first-integrated-optical-io-chiplet.html

SDM4 MCF MSA Addressing Fiber Challenges

SDM4 MCF MSA

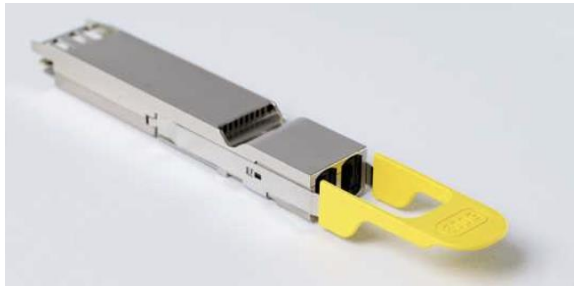


AFL CORNING SUMITOMO ELECTRIC TERAHOP

- **SDM4 MCF MSA** Launched (3/11/2026)
  - Optimized for short-reach, high-density O-band
- **Release 1.0** published on 9/17/2026
  - 4-core (2x2) enhance 4x capacity per fiber strand
  - Reduction of fiber count, weights, space, terminations
  - MCF fiber and MCF transceivers open demo



Source: Corning, OFC 2026 Demo



Source: TeraHop 1.6T MCF TRx



ECOC 2026 OIF Demo with TeraHop 800G-MCF



OFC 2026 TeraHop-Corning Joint Demo

SDM4 MCF for AI Scale Up

- "Slow and wide" optics gained industry attention as potential low power, low latency solutions for scale up
  - e.g., with 50G-NRZ modulation per lane, PAM4 DSP elimination greatly reduce power, and latency. However, 4x more channels needed.
- Options to gain back fiber efficiency:
  - Option #1: 4-wave DWDM per fiber to get to 200Gbps per fiber
  - Option #2: 4-core MCF with 1310nm to get to 200Gbps per fiber

	4-wave DWDM	4-Core MCF
Laser required	DWDM laser	~ 1310nm laser
Modulators	MRM	Modulator type agnostic
Optical MUX	Required (~3dB loss)	No WDM Mux
Power	Higher with DWDM laser and cooling/locking	Low power
Cost	Higher from high power DWDM laser, cooling, sensitive control	Lower cost, assuming 4-core MCF ~ 4x SMF

Call to Action

- AI infra drive demanding optical interconnect solutions
  - Hi capacity, Hi density, Lower power, High reliability/availability and supply resiliency
- Open optical interconnect solutions keeping up with Innovation
- Please join the Open optics community:
  - XPO MSA: https://xpomsa.com/
  - Open CPX MSA: https://www.opencpxmsa.org/
  - SDM4 MCF MSA: Contact AFL, Corning, Sumitomo, TeraHop (Terahop.com)







Thank You!

